Algebra II with Trigonometry Chapter 11.4

Olympiad Solutions (Gemini)

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Solutions

1. An ellipse in the coordinate plane has its foci at $(3, 4)$ and $(-5, 12)$. If the ellipse passes through the origin, what is the length of its major axis?

Solution: By the geometric definition of an ellipse, the sum of the distances from any point on the ellipse to the two foci is a constant equal to the length of the major axis, $2a$. Since the origin $(0, 0)$ lies on the ellipse, we simply compute the sum of the distances from the origin to the given foci $F_1(3, 4)$ and $F_2(-5, 12)$:

$$d_1 = \sqrt{3^2 + 4^2} = 5$$

$$d_2 = \sqrt{(-5)^2 + 12^2} = 13$$

The length of the major axis is $2a = d_1 + d_2 = 5 + 13 = 18$.

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2. The graph of the equation $x^2 - y^2 - 4x + 6y - 5 = 0$ and the line $x = 10$ enclose a bounded region in the coordinate plane. What is the area of this region?

Solution: First, we complete the square for both x and y to identify the given graph:

$$(x^2 - 4x + 4) - (y^2 - 6y + 9) = 5 + 4 - 9$$

$$(x - 2)^2 - (y - 3)^2 = 0$$

Factoring this difference of squares yields $((x - 2) - (y - 3))((x - 2) + (y - 3)) = 0$, which represents a degenerate hyperbola composed of two intersecting lines:

$$y = x + 1 \quad \text{and} \quad y = -x + 5$$

These lines intersect at $(2, 3)$. The region bounded by these two lines and the vertical line $x = 10$ forms a triangle. To find the base of the triangle on $x = 10$, we substitute $x = 10$ into the equations of the lines to find the vertices:

$$y_1 = 10 + 1 = 11$$

$$y_2 = -10 + 5 = -5$$

The length of the base is $11 - (-5) = 16$. The altitude of the triangle from the vertex $(2, 3)$ to the line $x = 10$ is $\Delta x = 10 - 2 = 8$. The area of the region is:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{altitude} = \frac{1}{2}(16)(8) = 64$$

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3. A parabola has its focus at $(0, 7)$ and its directrix is the horizontal line $y = -7$. If a point P on this parabola is at a distance of 25 from the focus, what is the shortest distance from P to the x-axis?

Solution: By the geometric definition of a parabola, every point $P(x, y)$ on it is equidistant from the focus and the directrix. If the distance from P to the focus is 25, then the perpendicular distance from P to the directrix $y = -7$ is also 25. Because the focus $(0, 7)$ is situated above the directrix, the parabola opens upwards and lies entirely in the upper half-plane $y \geq 0$. The distance from $P(x, y)$ to the directrix $y = -7$ is given by $y - (-7) = y + 7$.

$$y + 7 = 25 \implies y = 18$$

The y-coordinate of P is 18, meaning its shortest distance to the x-axis is simply 18.

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4. Consider the hyperbola given by the equation $3x^2 - y^2 - 12x + 10y - 100 = 0$. What is the degree measure of the acute angle between its asymptotes?

Solution: The slopes of a hyperbola's asymptotes are determined entirely by the highest-degree terms of its equation. Setting the quadratic homogeneous part to zero gives:

$$3x^2 - y^2 = 0 \implies y^2 = 3x^2 \implies y = \pm\sqrt{3}x$$

The asymptotes are parallel to the lines $y = \sqrt{3}x$ and $y = -\sqrt{3}x$. The slopes $\sqrt{3}$ and $-\sqrt{3}$ correspond to inclination angles of 60° and -60° (or 120°) with the positive x-axis. The angle between these two lines is $120^\circ - 60^\circ = 60^\circ$. Because $60^\circ < 90^\circ$, the degree measure of the acute angle between the asymptotes is 60°.

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5. An ellipse has its foci at $(3, -4)$ and $(3, 4)$, and it passes through the origin. What is the maximum y-coordinate of any point on this ellipse?

Solution: Because the foci $F_1(3, -4)$ and $F_2(3, 4)$ share the same x-coordinate, the major axis is vertical and is centered at the midpoint of the foci, $C(3, 0)$. The length of the semi-major axis, a, can be found using the origin $(0, 0)$, which lies on the ellipse. The sum of the distances to the foci equals $2a$:

$$2a = \sqrt{3^2 + (-4)^2} + \sqrt{3^2 + 4^2} = 5 + 5 = 10 \implies a = 5$$

The maximum y-coordinate of the ellipse is reached at its upper vertex. Starting from the center $C(3, 0)$ and moving a distance of $a = 5$ upwards along the major axis, we find the upper vertex at:

$$y_{\text{max}} = 0 + 5 = 5$$

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6. For what real value of k does the graph of the equation $x^2 + 4y^2 - 10x + 24y + k = 0$ degenerate into a single point?

Solution: We complete the square for both x and y to rewrite the equation in standard form:

$$(x^2 - 10x) + 4(y^2 + 6y) = -k$$

$$(x^2 - 10x + 25) + 4(y^2 + 6y + 9) = -k + 25 + 36$$

$$(x - 5)^2 + 4(y + 3)^2 = 61 - k$$

The left-hand side is a sum of non-negative terms. For the graph to be a single point (specifically, the point $(5, -3)$), the "radius squared" term on the right-hand side must be exactly zero.

$$61 - k = 0 \implies k = 61$$